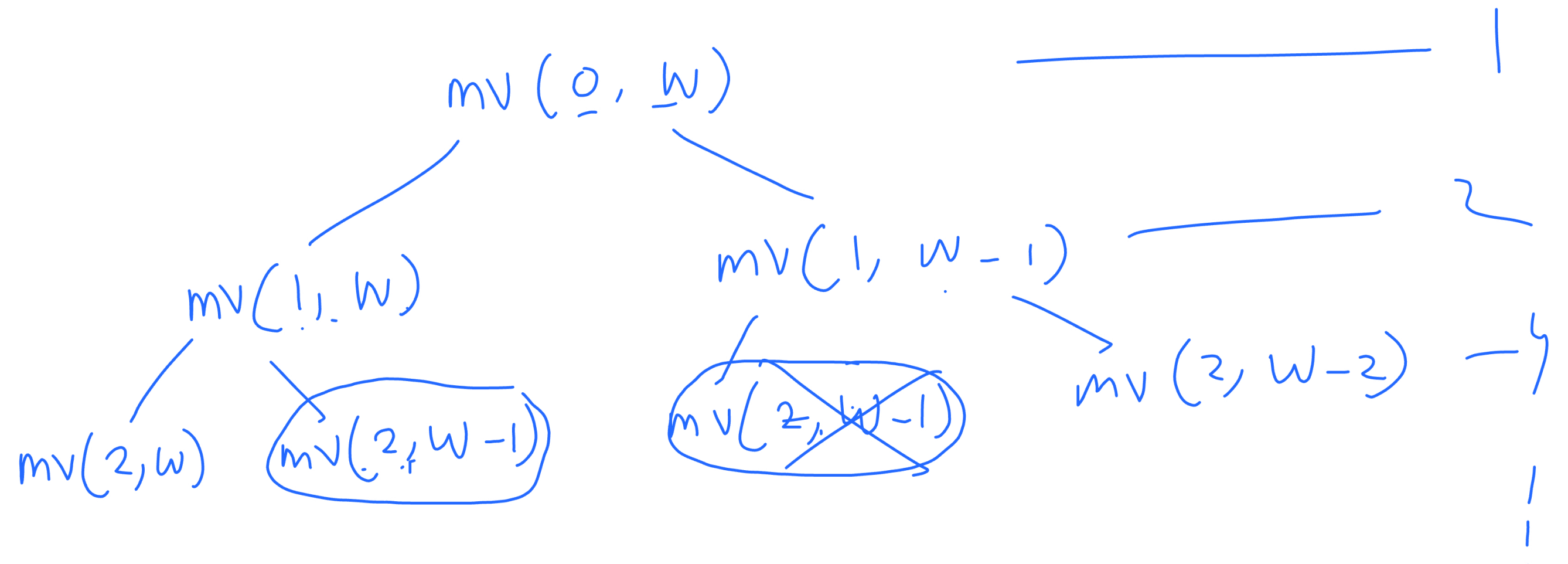

mv (0, W)
mv (1, W)
mv (1, W - 1)
mv (2, W)
mv (2, W - 1)
~~mv (2, W - 1)~~
mv (2, W - 2)
1
2
4

(i, w) → result

i: 0 → n-1
W: w → 0

n * w

Edit Dist.

A = ab ad
B = ab ac

O/P: 1

A = anshuman
B = antihuman

O/P: an~~s~~human → 2
ti

~~anshuman~~
~~antshuman~~

$$ED(anshuman, antihuman)$$

↓

$$ED(nshuman, ntihuman)$$

↓

$$ED(shuman, tihuman)$$

insert: $1 + ED(shuman, ihuman)$

del: $1 + ED(human, tihuman)$

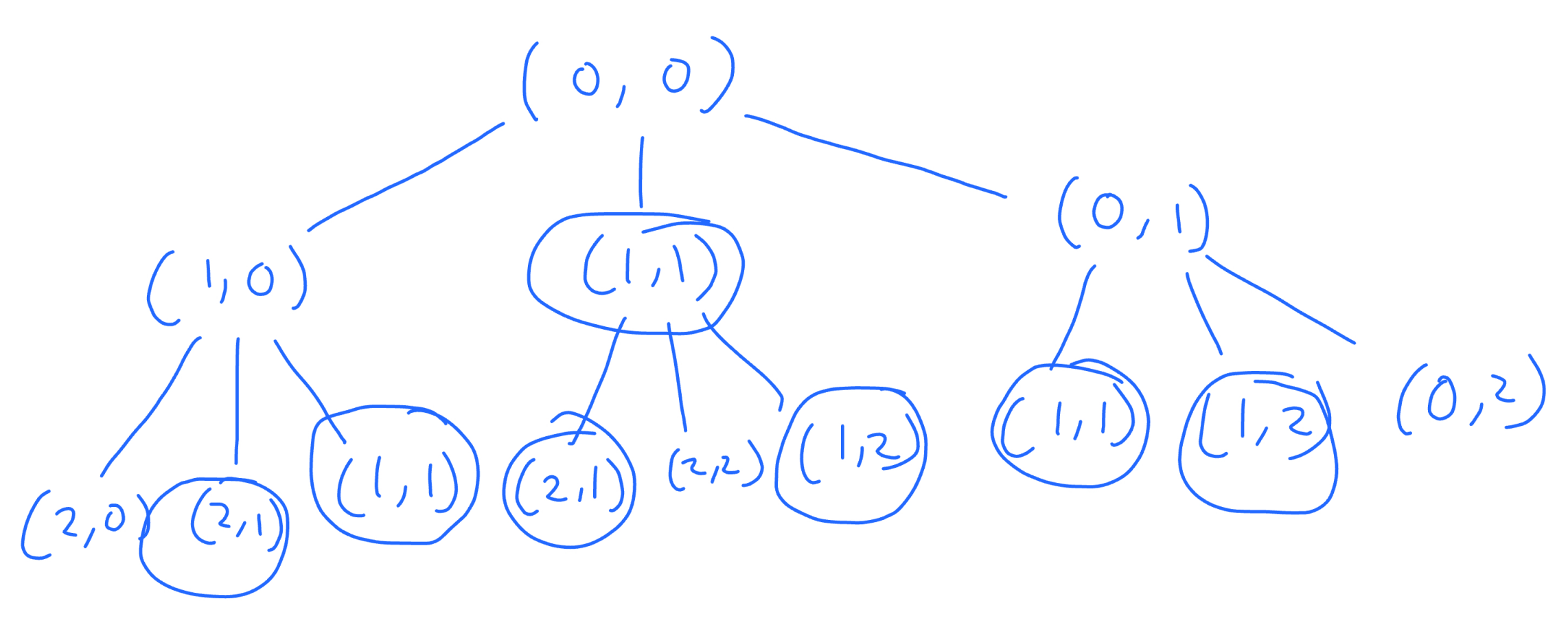
replace: $1 + ED(human, ihuman)$

$$ED(i, j) = \begin{cases} ED(i+1, j+1) & \text{if } A[i] == B[j] \\ 1 + \min\begin{pmatrix} ED(i, j+1), \\ ED(i+1, j), \\ ED(i+1, j+1) \end{pmatrix} & \text{otherwise} \end{cases}$$

ED("", "abc") = 3

ED("", anshuman)   0 1 2 3 4 5 6 7   8 - j

ED(abc, "") = 3.


(0, 0)
(1, 0)
(1, 1)
(0, 1)
(2, 0)
(2, 1)
(1, 1)
(2, 1)
(2, 2)
(1, 2)
(1, 1)
(1, 2)
(0, 2)